DATASET SUMMARY

Students Performance in Exams Dataset

Materi: Eksplorasi Awal Dataset — Hari Kamis
Tools: Python (Pandas, Matplotlib)

1. Screenshot Dataset

df.head() — 5 baris pertama

```
df.head()
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75

df.tail() — 5 baris terakhir

```
df.tail()
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
995	female	group E	master's degree	standard	completed	88	99	95
996	male	group C	high school	free/reduced	none	62	55	55
997	female	group C	high school	free/reduced	completed	59	71	65
998	female	group D	some college	standard	completed	68	78	77
999	female	group D	some college	free/reduced	none	77	86	86

2. Info Dataset

df.info()

```
df.describe()
```

	math score	reading score	writing score
count	1000.00000	1000.000000	1000.000000
mean	66.08900	69.169000	68.054000
std	15.16308	14.600192	15.195657
min	0.00000	17.000000	10.000000
25%	57.00000	59.000000	57.750000
50%	66.00000	70.000000	69.000000
75%	77.00000	79.000000	79.000000
max	100.00000	100.000000	100.000000

Keterangan	Nilai
Jumlah baris	1000
Jumlah kolom	8
Kolom numerik	math score, reading score, writing score
Kolom kategorikal	gender, race/ethnicity, parental level of education, lunch, test preparation course
Missing value	Tidak ada (0 di semua kolom)
Baris duplikat	Tidak ada (0)

3. Statistik Dasar (df.describe())

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column                       Non-Null Count  Dtype 
---  ------                       --------------  ----- 
 0   gender                       1000 non-null   object
 1   race/ethnicity               1000 non-null   object
 2   parental level of education  1000 non-null   object
 3   lunch                        1000 non-null   object
 4   test preparation course      1000 non-null   object
 5   math score                   1000 non-null   int64 
 6   reading score                1000 non-null   int64 
 7   writing score                1000 non-null   int64 
dtypes: int64(3), object(5)
memory usage: 62.6+ KB
```

Rata-rata nilai matematika (66.1), membaca (69.2), dan menulis (68.1) berada pada kisaran yang mirip, dengan nilai membaca sedikit lebih tinggi. Standar deviasi ketiganya berkisar 14–15 poin, menunjukkan variasi nilai antar siswa cukup lebar.

4. Missing Value & Data Unik

Jumlah Missing Value per Kolom

df.isnull().sum()

```
df.isnull().sum()
```

	0
gender	0
race/ethnicity	0
parental level of education	0
lunch	0
test preparation course	0
math score	0
reading score	0
writing score	0

dtype: int64

Kolom	Missing Value
gender	0
race/ethnicity	0
parental level of education	0
lunch	0
test preparation course	0
math score	0
reading score	0
writing score	0

Kesimpulan: Dataset ini bersih dari missing value dan data kosong — semua 1000 baris terisi lengkap di 8 kolom.

Jumlah Nilai Unik per Kolom

df.nunique()

```
df[df.isnull().any(axis=1)]
```

No entries Filter

index	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score

Show 25 per page
Like what you see? Visit the data table notebook to learn more about interactive tables.

Kolom	Jumlah Nilai Unik
gender	2

race/ethnicity	5
parental level of education	6
lunch	2
test preparation course	2
math score	81
reading score	72
writing score	77

5. Distribusi Kategori

value_counts() per kolom kategorikal

```
df.nunique()
```

	0
gender	2
race/ethnicity	5
parental level of education	6
lunch	2
test preparation course	2
math score	81
reading score	72
writing score	77

dtype: int64

Gender

Female 518 (51.8%), Male 482 (48.2%) — hampir seimbang.

Lunch

Standard 645 (64.5%), Free/Reduced 355 (35.5%).

Test Preparation Course

None 642 (64.2%), Completed 358 (35.8%) — mayoritas siswa belum mengikuti kursus persiapan.

6. Outlier Awal (Metode IQR)

```
df["gender"].value_counts()
```

gender	count
female	518
male	482

dtype: int64

Kolom	Batas Bawah	Batas Atas	Jumlah Outlier
Math Score	27	107	8
Reading Score	29	109	6
Writing Score	25.9	110.9	5

Outlier terdeteksi pada nilai yang sangat rendah (misalnya math score = 0), kemungkinan siswa yang tidak mengikuti ujian atau kesalahan input data, bukan outlier pada nilai tinggi.

7. Insight Awal

- Dataset bersih: tidak ada missing value maupun baris duplikat di 1000 data.
- Distribusi gender cukup seimbang (51.8% perempuan vs 48.2% laki-laki).
- Mayoritas siswa (64.2%) belum mengikuti test preparation course.
- Nilai reading score memiliki rata-rata tertinggi (69.2) dibanding math dan writing.
- Terdapat beberapa outlier nilai sangat rendah (termasuk math score = 0) yang perlu diperiksa lebih lanjut — apakah data valid atau perlu dibersihkan.
- Kategori race/ethnicity didominasi Group C (319 siswa) dan Group D (262 siswa).
- Variasi nilai (std ±14-15) menunjukkan performa siswa cukup beragam, membuka peluang analisis lanjutan terhadap faktor-faktor yang memengaruhi nilai.